FPS-3000 SBC — VersaBus address map

Derived from FPS3K_U11_U12_JOIN.bin + emulator measurement. Supersedes the 2026-07 edition; see PDF_ERRATA.md for what changed.

Windows

range	device	contents
$FF0000-$FF001F	AP I/F	command reg + bulk data port (see below)
$FF0040-$FF00BF	AP I/F	four 32-byte channel windows, $FF0040+$20*N
$FF0200-$FF021B	XLTR	mode / select / counter / data / status / IRQ mask
$FF0230-$FF025F	XLTR	three MC68153-style BIMs, 4 channels each
$70001C/$700020	mailbox	host->SBC / SBC->host, one 32-bit word each way
$400000-$4FFFFF	chassis	paged memory window; MODE2 = addr bits 20-31

AP I/F command window $FF0000-$FF001F

offset	addr	dir	role
+$00	$FF0000	R/W	command reg; $8004/$8005 out. Reads <=0 mean END OF STREAM
+$04	$FF0004	R	bit 0 = port ready; polled before every transfer (F04B22)
+$08	$FF0008	R/W	BULK DATA PORT, bidirectional - three modes, below
+$0E	$FF000E	W	panel command / argument staging

$FF0008 mode is chosen by the latched opcode $E5C and bit 5 of the chassis response byte:

$E5C / bit5	direction	content
$00 / 0	in	ASCII S-record text, two characters per 16-bit word
$28 / -	in	binary, word -> staging buffer at $E58
$00 / 1	out	binary, staging buffer -> port (this is the WCS upload)

All three use the same handshake: STATUS_IRQ <- $400, poll bit 15, STATUS_IRQ <- 0.

AP I/F channel windows $FF0040 + $20*N   [CORRECTED]

offset	ch 1	ch 2	ch 3	ch 4	role
+$04	$FF0044	$FF0064	$FF0084	$FF00A4	write port
+$08	$FF0048	$FF0068	$FF0088	$FF00A8	32-bit data, HIGH half
+$0A	$FF004A	$FF006A	$FF008A	$FF00AA	32-bit data, LOW half
+$0E	$FF004E	$FF006E	$FF008E	$FF00AE	command/trigger ($8000 fires)

+$08 and +$0A are one 32-bit register, not data+status: BLK_XFR (F05B0E) reads both each pass; TCBXP1I writes $0000001B across the pair at F07EC6 then $8000 to +$0E. Eight Am29705 16x4 two-port RAMs give exactly 32 bits. $FF0048 is NEVER READ by the ROM.

XLTR control file $FF0200-$FF021B

addr	name	established behaviour
$FF0200	MODE0	bits 0-7 chassis response byte; b10 ack, b11 valid. Low byte reads 0 when idle
$FF0202	MODE1	b7 busy (tested 15x), b12 enable (set 8x), b14 control (cleared 12x), b15 arms chassis op
$FF0204	CHANNEL_SELECT	BIDIRECTIONAL: chassis presents args, SBC returns results from $E74
$FF020C	COUNTER	operational value $4 (7 sites); $1/$FF are boot-diagnostic only; readable
$FF0210	MODE2	chassis page select = address bits 20-31 of the $400000 window
$FF0214	DATA_LO	used by phase $1900 chassis-memory data test
$FF0216	DATA_HI	mode/page reg, NOT a command reg; gates $400000 access. Resting value $C0
$FF0218	STATUS_IRQ	arm $400 / poll b15 / clear 0. b4 = BIM POPULATION (16 vs 24 regs)
$FF021A	IRQ_MASK	reads $FFF. Bits 5,4,3,2 = XP channels 1,2,3,4 (PanelErrorMaskTable F05C4C)

FPS-3000 SBC — interrupts, protocol, bring-up

Companion to page 1. Every claim here is either read from the ROM or measured in the emulator.

BIM channels $FF0230-$FF025F   (CR n at +0/+2/+4/+6, VR n at +8/+A/+C/+E)

BIM.ch	CR	value	VR	vector	handler	owner
0.0	$FF0230	$5E	$FF0238	$41 -> $104	F04930	TCBRDHC
1.2	$FF0244	$5F	$FF024C	$45 -> $114	F07EE6	TCBXP1I
1.3	$FF0246	$5F	$FF024E	$46 -> $118	F074E6	TCBXP2I
2.0	$FF0250	$5F	$FF0258	$47 -> $11C	F06AE6	TCBXP3I
2.1	$FF0252	$5F	$FF025A	$48 -> $120	F060CE	TCBXP4I
2.2	$FF0254	$5F	$FF025C	$4A -> $128	F05DD6	TCBIO1I

CR bits: 0-2 = IRQ level (0 disables), 4 = IRE enable, 5 = IRAC auto-clear (off), 7 = Flag. $4F = $5F with IRE cleared - PanelSendAndWait writes it to suppress a channel's interrupt during a transfer. $4F has NO connection to $FF004A.

Chassis -> SBC command protocol (responses arrive on BIM0 ch0)

code	operation	argument (CHANNEL_SELECT)	result
$0	latch opcode, execute	opcode 0..$10, or $28 = bulk	-
$1 / $41	set address low / high	address half -> $E58/$E5A	-
$2 / $42	set count low / high	count half -> $E64/$E66	-
$3/$43/$63	chassis mem write/hi/read	data half -> $E70/$E72	$E70
$5	arg 0: report AC count	0, or channel number	$105E
$7	disable BIM0 ch0 IRQ	-	-
$9	set data parameter	data half -> $E68/$E6A	-
$B	report $10010	-	$E74
$F	return from interrupt	-	-

Bit 7 of the response byte selects the dispatcher (0 = the 16-entry table at F05102). Bit 6 selects the half for the loaders. Bit 5 selects direction for code $3 and for the bulk transfer - opposite senses, do not generalise.

Microcode upload — verified end to end in one session

host ASCII S-records -> $FF0008 -> SRecordDataHandler -> staging $10010+ -> $FF0008 -> XLTR -> UNIV FMT -> WCS

S-RECORD ADDRESSES ARE OFFSETS. The handler computes a1 = $10 + record_address + $10000 (seed at F051A2, add at F051DC) and range-checks the RESULT to $10000-$1FFFF. Usable record range $0000-$FFEF -> lands $10010-$1FFFF.
There is NO SBC-side WCS bank select: the outbound loop carries only source address and count, no destination.
The chassis is the bus master and places the data; the SBC is a slave conduit and never asserts a transfer request.

Bring-up: the self-test phase beacon

Every power-on self-test writes phase<<8 | subtest to $FF0204. Scope or read that register during reset and the last value before a hang is the failing subtest - no debugger, no serial port needed.

beacon	suspect	beacon	suspect
$01xx	CPU flags/arithmetic	$1000	address-space boundary map
$02xx	$1FFF1 b6 <-> $F70019 b3	$11xx-$14xx	panel-bus IRQs and vectors
$0300	ROM decode	$1500	CHANNEL_SELECT read/write
$0400	RAM address decode	$1600	XLTR file / BIM population bit
$0500	(status & $3F31) == $3F11	$17xx-$18xx	DATA_HI page gating
$0600	VMOD pattern round-trip	$1900	chassis memory data lines
$0700	short-I/O must bus-error	$1A00	AP I/F window
$0800	I/O channel	$2000	RAM address uniqueness
$0900	MC6840 PTM + $1FFF1 b7	$22xx-$23xx	chassis bulk / A14 decode

Six of Motorola's twelve board-status inputs are USER-DEFINED (Table 1, M68KVM02-3 p.2-59). $F70019 bits 1-5 are FPS's own wiring and will never appear in a datasheet - infer from firmware, or trace on the board.